A LOCAL RING WHOSE M-ADIC COMPLETION IS LOCAL BUT NOT COMPLETE AND WHOSE MAXIMAL IDEAL IS STRICTLY LARGER THAN THE EXPANDED ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof adds only the localization bridge ('the completion of the localization R_m is equal to the completion of R'), the author's own one-sentence assertion; the three verifications live in the context span and are complete there.

Editorial note. The label lemma-noncomplete-completion names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 525-651: for R = k[x_1, x_2, ...] with m = (x_1, x_2, ...), the completion R^ = lim R/m^n is local with maximal ideal m' = Ker(R^ -> R/m); the element f = x_1 + x_2^2 + x_3^3 + ... lies in m' but not in m R^ (falsifying (1)); Algebra lemma-hathat with n=1 then falsifies (3); the Mittag-Leffler exact sequences reduce m'-completeness to K_n = (m')^n, and the explicit z = sum x_i^d with d prime to char(k) (vanishing locus of partial derivatives one point vs Leibniz rule) shows K_2 != (m')^2, falsifying (2) - the whole argument from an unpublished note of de Smit and Lenstra, complete in the section
- Algebra, lemma-characterize-local-ring, lemma-hathat, lemma-Mittag-Leffler

Source reference. Original lemma 03CA: sources/stacks-chow/tags.txt, source lines 4377–4377. Prerequisite pointer invoked at 734; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00E9: sources/stacks-chow/tags.txt, source lines 526–526. Prerequisite pointer invoked at 682; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0318: sources/stacks-chow/tags.txt, source lines 3975–3975. Prerequisite pointer invoked at 713; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 07E7: sources/stacks-chow/tags.txt, source lines 9437–9437. Prerequisite pointer invoked at 685; it names a standard fact of a sibling

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chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let R be a ring and let $\mathfrak{m}$ be a maximal ideal. Consider the completion

$$R^\wedge = \lim R/\mathfrak{m}^n.$$

Note that $R^\wedge$ is a local ring with maximal ideal $\mathfrak{m}' = \mathrm{Ker}(R^\wedge \to R/\mathfrak{m})$. Namely, if $x = (x_n) \in R^\wedge$ is not in $\mathfrak{m}'$, then $y = (x_n^{-1}) \in R^\wedge$ satisfies $xy = 1$, whence $R^\wedge$ is local by Algebra, Lemma 00E9. Now it is always true that $R^\wedge$ complete in its limit topology (see the discussion in More on Algebra, Section 07E7). But beyond that, we have the following questions:

(1) Is it true that $\mathfrak{m}R^\wedge = \mathfrak{m}'$?
(2) Is $R^\wedge$ viewed as an $R^\wedge$-module $\mathfrak{m}'$-adically complete?
(3) Is $R^\wedge$ viewed as an R-module $\mathfrak{m}$-adically complete?

It turns out that these questions all have a negative answer.

Let k be a field, $R = k[x_1, x_2, x_3, \ldots]$, and $\mathfrak{m} = (x_1, x_2, x_3, \ldots)$. We will think of an element f of $R^\wedge$ as a (possibly) infinite sum

$$f = \sum a_I x^I$$

(using multi-index notation) such that for each $d \geq 0$ there are only finitely many nonzero a_I for $|I| = d$. The maximal ideal $\mathfrak{m}' \subset R^\wedge$ is the collection of f with zero constant term. In particular, the element

$$f = x_1 + x_2^2 + x_3^3 + \ldots$$

is in $\mathfrak{m}'$ but not in $\mathfrak{m}R^\wedge$ which shows that (1) is false in this example. However, if (1) is false, then (3) is necessarily false because $\mathfrak{m}' = \mathrm{Ker}(R^\wedge \to R/\mathfrak{m})$ and we can apply Algebra, Lemma 0318 with $n = 1$.

To finish we prove that $R^\wedge$ is not $\mathfrak{m}'$-adically complete. For $n \geq 1$ let $K_n = \mathrm{Ker}(R^\wedge \to R/\mathfrak{m}^n)$. Then we have short exact sequences

$$0 \to K_n/(\mathfrak{m}')^n \to R^\wedge/(\mathfrak{m}')^n \to R/\mathfrak{m}^n \to 0$$

The projection map $R^\wedge \to R/\mathfrak{m}^{n+1}$ sends $(\mathfrak{m}')^n$ onto $\mathfrak{m}^n/\mathfrak{m}^{n+1}$. It follows that $K_{n+1} \to K_n/(\mathfrak{m}')^n$ is surjective. Hence the inverse system $(K_n/(\mathfrak{m}')^n)$ has surjective transition maps and taking inverse limits we obtain an exact sequence

$$0 \to \lim K_n/(\mathfrak{m}')^n \to \lim R^\wedge/(\mathfrak{m}')^n \to \lim R/\mathfrak{m}^n \to 0$$

by Algebra, Lemma 03CA. Thus we see that $R^\wedge$ is complete with respect to $\mathfrak{m}'$ if and only if $K_n = (\mathfrak{m}')^n$ for all $n \geq 1$.

To show that $R^\wedge$ is not $\mathfrak{m}'$-adically complete in our example we show that $K_2 = \mathrm{Ker}(R^\wedge \to R/\mathfrak{m}^2)$ is not equal to $(\mathfrak{m}')^2$. Note that an element of $(\mathfrak{m}')^2$ can be written as a finite sum

$$\sum\nolimits_{i=1,\ldots,t} f_i g_i \tag{0.0.1}$$

with $f_i, g_i \in R^\wedge$ having vanishing constant terms. To get an example we are going to choose an $z \in K_2$ of the form

$$z = z_1 + z_2 + z_3 + \ldots$$

with the following properties

(1) there exist sequences $1 < d_1 < d_2 < d_3 < \ldots$ and $0 < n_1 < n_2 < n_3 < \ldots$ such that $z_i \in k[x_{n_i}, x_{n_i+1}, \ldots, x_{n_{i+1}-1}]$ homogeneous of degree d_i, and
(2) in the ring $k[[x_{n_i}, x_{n_i+1}, \ldots, x_{n_{i+1}-1}]]$ the element z_i cannot be written as a sum (0.0.1) with $t \leq i$.

Clearly this implies that z is not in $(\mathfrak{m}')^2$ because the image of the relation (0.0.1) in the ring $k[[x_{n_i}, x_{n_i+1}, \ldots, x_{n_{i+1}-1}]]$ for i large enough would produce a contradiction. Hence it suffices to prove that for all $t > 0$ there exists a $d \gg 0$ and an integer n such that we can find an homogeneous element $z \in k[x_1, \ldots, x_n]$ of degree d which cannot be written as a sum (0.0.1) for the given t in $k[[x_1, \ldots, x_n]]$. Take $n > 2t$ and any $d > 1$ prime to the characteristic of k and set $z = \sum_{i=1,\ldots,n} x_i^d$. Then the vanishing locus of the ideal

$$(\frac{\partial z}{\partial x_1}, \ldots, \frac{\partial z}{\partial x_n}) = (dx_1^{d-1}, \ldots, dx_n^{d-1})$$

consists of one point. On the other hand,

$$\frac{\partial(\sum_{i=1,\ldots,t} f_i g_i)}{\partial x_j} \in (f_1, \ldots, f_t, g_1, \ldots, g_t)$$

by the Leibniz rule and hence the vanishing locus of these derivatives contains at least

$$V(f_1, \ldots, f_t, g_1, \ldots, g_t) \subset \mathrm{Spec}(k[[x_1, \ldots, x_n]]).$$

Hence this is a contradiction as the dimension of $V(f_1, \ldots, f_t, g_1, \ldots, g_t)$ is at least $n - 2t \geq 1$.

Lemma 0.1. *There exists a local ring R and a maximal ideal $\mathfrak{m}$ such that the completion $R^\wedge$ of R with respect to $\mathfrak{m}$ has the following properties*

(1) *$R^\wedge$ is local, but its maximal ideal is not equal to $\mathfrak{m}R^\wedge$,*
(2) *$R^\wedge$ is not a complete local ring, and*
(3) *$R^\wedge$ is not $\mathfrak{m}$-adically complete as an R-module.*

Proof. This follows from the discussion above as (with $R = k[x_1, x_2, x_3, \ldots]$) the completion of the localization $R_\mathfrak{m}$ is equal to the completion of R. □